

When Butterfly Meets Photoacoustic Effect

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Abstract

The photoacoustic effect converts light into sound, bridging optical excitation and acoustic generation. However, existing photoacoustic systems remain largely confined to specialized applications because the generated signals are weak and typically require expensive lasers and delicate materials. In this paper, we demonstrate for the first time that natural butterfly wings can function as highly effective photoacoustic materials, enabling strong and practical photoacoustic generation using an ordinary LED. The key principle is that butterfly wings are not only excellent light absorbers but also support efficient light-heat-acoustic conversion, owing to their unique ultra-thin and lightweight membrane structures. In this work, we establish a model of the butterfly photoacoustic effect, providing a theoretical foundation for future research. We address critical technical challenges, including weak signal, as well as practical challenges such as environmental interference, realizing the first end-to-end butterfly photoacoustic system. Results show that our system achieves strong photoacoustic outputs exceeding 70 dB—comparable to a speaking voice—using off-the-shelf LEDs and naturally deceased butterfly specimens. We demonstrate the application potential of the proposed butterfly photoacoustic effect by using the butterfly as a bio-speaker for audible information delivery and for fine-grained ultrasound sensing. Our system harnesses this intriguing cross-energy phenomenon for practical applications and opens new opportunities for bio-inspired designs.

CCS Concepts

• Human-centered computing → Ubiquitous and mobile computing systems and tools.

Keywords

bio-inspired photoacoustics, butterfly wings, bio-triggered wireless sensing



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1 Introduction

In the late 1870s, Alexander Graham Bell invented not only the now-famous electromagnetic telephone, but also the lesser-known photophone based on the photoacoustic effect [39]. The photoacoustic effect refers to the phenomenon in which light with time-varying intensity illuminates a material, inducing periodic heating that causes thermal expansion and contraction, thereby generating acoustic waves. By carefully modulating the light source, the illuminated object can act as a receiver and reproduce sound signals. Although the photophone was eventually overshadowed due to energy inefficiency during multiple stages of conversion, the underlying photoacoustic effect reveals a broader scientific principle: it bridges two distinct physical carriers—*light*, an electromagnetic wave, and *sound*, a mechanical pressure wave. This coupling also presents promising opportunities to jointly exploit the complementary physical properties of light and acoustic signals. A representative example is biomedical photoacoustic imaging [3, 8, 51], which leverages the high spatial resolution of optical excitation and the deeper tissue penetration of acoustic signals.

However, the fundamental limitation that led to the failure of the photophone still persists and continues to hinder the broader deployment of the photoacoustic effect: the generated acoustic signals are inherently weak due to multiple stages of energy conversion. As a result, existing systems typically require very short operating distances and dedicated light transmitters and acoustic receivers, which are often bulky and prohibitively expensive. Over the past two decades, most studies [18, 19, 21, 40, 53] have sought to enhance the intensity of the generated photoacoustic signals. Given that modern light sources, such as LEDs, can already provide sufficiently high illumination intensity, the primary bottleneck lies in the energy conversion efficiency at the illuminated object. Accordingly, prior efforts have either (i)

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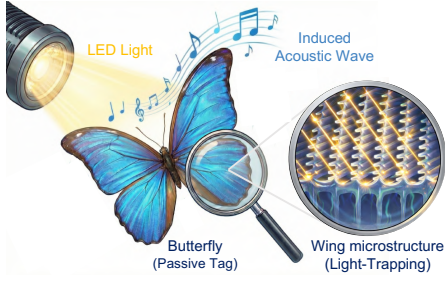


Figure 1: Butterfly photoacoustic system overview.

designed dedicated optical structures such as reflective cavities [18, 53] to enhance optical absorption, or (ii) developed specialized materials (e.g., nanocomposites) that more efficiently convert light into heat [19, 21, 40]. Although promising, their reliance on delicate fabrication processes, such as nanofabrication [22, 31], has so far confined photoacoustic technologies to specialized and costly settings.

This naturally raises a compelling question: *Can we discover a fundamentally different pathway to overcome the key barrier hindering the widespread adoption of the photoacoustic effect and bring photoacoustic technologies into everyday life?* To explore this possibility, we turn to nature, an unparalleled source of inspiration. Butterfly wings, known for their exceptional light-absorbing capabilities, have attracted considerable attention from the photonics community [9, 15, 46]. The natural nanostructure of butterfly wings traps incident light through repeated reflection and scattering (as shown in Figure 1), enabling the light to be absorbed multiple times across the wing. As a result, this natural structure can achieve light-harvesting performance that surpasses conventional engineered (artificial) designs. Because light absorption efficiency critically influences the photoacoustic effect, butterfly wings offer a compelling natural inspiration for bio-inspired photoacoustic designs. Furthermore, they are readily accessible through established marketplaces for decorative and jewelry products [5] and are biodegradable and environmentally friendly [37]. Taken together, these properties make butterfly wings, whose structures and materials have been shaped by millions of years of natural evolution, a promising candidate for enhancement of photoacoustic effect. However, several challenges still remain to be addressed.

The first challenge arises from *the lack of clear understanding of the mechanisms governing the full energy-conversion chain in the butterfly photoacoustic effect*. Unlike simple optical energy absorption in existing butterfly studies, the photoacoustic effect involves a multi-stage light–heat–acoustic conversion process. Strong optical absorption of butterfly wings does not necessarily translate into a strong photoacoustic response. For example, the well-known Blue Morpho

butterfly [9, 12] (also cf. Figure 2(a)–(b)), whose optical absorption capability is exceptional among butterfly species, does not outperform other species in photoacoustic performance based on our measurements. This observation shows that the role of butterfly wing nanostructures in the photoacoustic process remains insufficiently understood.

To address this challenge, we develop a multi-stage model that characterizes the light–heat–acoustic conversion process within butterfly wing nanostructures and validate it experimentally. The model reveals that, in addition to light absorption efficiency, other factors such as heat capacity, mass, size, and damping characteristics of butterfly wings also play important roles in determining the final acoustic emission. It further explains that the relatively weak photoacoustic performance of the Blue Morpho butterfly stems from its heavier and stiffer wings, which suppress thermally induced vibrations. More broadly, the model provides the theoretical foundation for subsequent optical excitation (transmitter) design and butterfly wing (tag) design.

The second challenge lies in *how to design transmitter-side optical modulation to induce strong and stable photoacoustic signals*. An intuitive approach to driving the light source (e.g., an LED) is to modulate it with a single-tone sinusoidal signal, similar to that used in conventional acoustic systems. However, the resulting photoacoustic output is weak, reaching only around 40 dB, which is audible but still insufficient for many practical applications. Moreover, we observe that the frequency response of the butterfly wing’s photoacoustic effect is not flat; in other words, the same light intensity at different frequencies can produce dramatically different acoustic amplitudes. This non-uniform response leads to a significant issue: the intensity of the generated photoacoustic signal fluctuates over time as the acoustic frequency continuously varies (e.g., during music playback).

Guided by our model, we first find that the amplitude of the photoacoustic signal depends not only on the light intensity but also on the transition slope of the light signal. For the same intensity, a steeper transition produces faster thermoelastic expansion, thereby inducing stronger acoustic emission. Thus, we replace the sinusoidal modulation with square-wave modulation to achieve sharper transitions, boosting the photoacoustic output from 40 dB to around 60 dB. We further find that strong photoacoustic effect depends not only on sharp transitions, but also on proper light-ON and light-OFF durations within each cycle. If these durations are too short, the butterfly wing does not have enough time to heat and cool sufficiently, limiting the photoacoustic output. By incorporating these insights into our light ON-OFF design, we successfully increased the photoacoustic output to 70 dB—a level comparable to normal conversation and sufficient for our target applications.

To investigate the source of the fluctuating signal amplitude, we find that the audio frequency is determined by the duration of the ON–OFF cycles: shorter cycles correspond to higher frequencies. While adjusting the cycle duration controls the audio frequency, it also affects signal strength, resulting in fluctuating amplitudes. To address this issue, we adopt a fixed high-frequency carrier (e.g., 20 kHz, in the inaudible range for humans), which ensures a stable acoustic signal amplitude. We modulate the acoustic signal onto the carrier as envelopes and adjust the envelope period to control the signal frequency. The interesting fact is that, although the acoustic output contains both a high-frequency component (carrier) and a low-frequency component (envelope), the high-frequency ultrasound is naturally filtered by the human ear, so no demodulation is required to separate them.

The third challenge is *how to design a practical butterfly wing tag that improves both the strength and quality of the emitted acoustic signals*? While there are tens of thousands of butterfly species in the wild, only a few dozen are widely raised commercially, making them potential candidates for our system. Besides, different wing sizes and structural regions exhibit distinct mechanical and photothermal properties, further complicating the design choices. Moreover, even though our transmitter-side design enhances signal strength and stability, the photoacoustic signals generated by butterfly wings still contain irregular nonlinear components that are difficult to model or eliminate. We find that the irregular components mainly come from two sources. First, the butterfly wing is ultra-light, making it highly sensitive to external perturbations such as airflow. Second, this ultra-light membrane is mechanically non-uniform, allowing multiple vibration modes to be excited and coupled, which produces irregular spectral components beyond simple harmonics.

To address this challenge, we apply both the proposed model and experiments to identify appropriate butterfly species, and find that strong photoacoustic effects can be achieved using cheap, widely available, commercially bred species. We then identify suitable wing regions that achieve the best overall performance across multiple stages of the photoacoustic conversion process. To suppress those irregular signal components, we stabilize the wing by sandwiching it between two ultra-thin transparent films made of low-cost and thermally stable polyethylene (PE) [38]. This design gently and uniformly improves the mechanical stability of the wing while minimally affecting the photoacoustic process. With PE encapsulation, the irregular nonlinear spectral components are significantly suppressed, while the photoacoustic signal strength decreases by only 1.7%.

With these designs, we develop ButterflyEffect, an end-to-end butterfly photoacoustic system that uses an off-the-shelf LED (\$2) together with farmed, naturally deceased butterfly

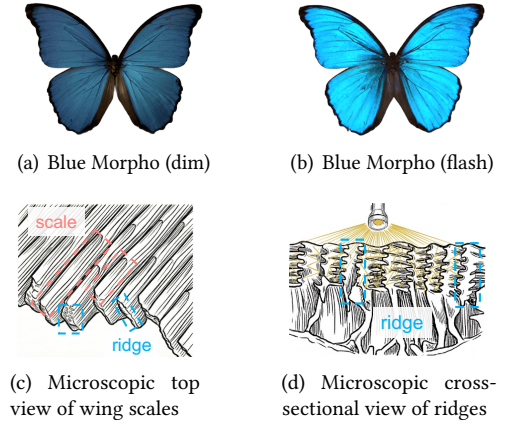


Figure 2: Optical appearance of a butterfly wing under different illumination and its nanostructures.

specimens (each under \$0.5) to achieve a pronounced photoacoustic effect, reaching sound pressure levels of up to 73 dB. We further demonstrate the real-life potential of butterfly photoacoustic effect through two case studies: (i) a butterfly loudspeaker for human-perceivable information delivery in the audible band, enabling applications such as interactive audio guides in museums—requiring no electronic devices on the user, only a single butterfly and (ii) a sensing platform capable of tracking both large motions, such as hand gestures, and small motions, such as respiration, in the ultrasonic band. Our main contributions are as follows:

- To the best of our knowledge, this is the first work to identify and show the remarkable photoacoustic performance of butterfly wings. We develop a theoretical model of the butterfly photoacoustic effect and validate it experimentally, providing theoretical insights for future studies.
- We design and implement an end-to-end butterfly photoacoustic system, incorporating solutions to address challenges such as weak signals, uneven response, and practical environmental interference. We enable a strong photoacoustic effect on inexpensive, portable, everyday hardware (e.g., ordinary LEDs) for the first time.
- We demonstrate the feasibility and potential of the proposed photoacoustic platform through two case studies: using the butterfly as a bio-speaker for audible information delivery, and employing it for fine-grained ultrasound sensing. We hope this system will inspire further bio-inspired designs and applications within our community.

2 Background

In this section, we describe the stages of photoacoustic energy conversion and the properties of butterfly wings.

Energy conversion in the photoacoustic effect. The photoacoustic effect generates acoustic waves when a material is illuminated by intensity-modulated light [39]. From the energy-conversion perspective, the light-heat-acoustic conversion can be decomposed into three stages:

Stage I (optical absorption): incident light is absorbed by the material according to its optical absorption efficiency.

Stage II (photothermal conversion): the absorbed optical energy is converted into heat, producing a temperature rise $\Delta T(t)$ governed by the material's thermal properties [14, 57].

Stage III (thermal-to-mechanical conversion): such temperature oscillations induce a thermoelastic effect in the material, generating vibrations that radiate acoustic waves.

Optical, thermal, and mechanical properties of butterfly wings. Butterfly wings are widely known for their hierarchical nanostructures (Figure 2(a)-(b)), which enable strong light absorption through multiple internal reflections. As shown in Figure 2(c)-(d), a butterfly wing contains many small scales [5], and each scale features ridge structures that repeatedly redirect light within the scales, thereby enhancing optical absorption. Thermally, wings are porous chitin-based membranes with abundant air cavities [50], which give them low heat capacity and rapid heat exchange with the surrounding air. Mechanically, the thin membrane and scale layers are supported by a flexible vein network [34], enabling low-mass vibrations under thermoelastic driving. Together, these observations suggest that butterfly wings are highly suitable for efficient photoacoustic generation.

3 System Overview

Figure 3 shows our butterfly photoacoustic system: a commodity LED serves as the optical transmitter, the butterfly wing acts as a passive photoacoustic tag (acoustic transmitter), and a commodity microphone or the human ear functions as the acoustic receiver. As summarized in Figure 4, the system consists of three key components: a stage-by-stage energy-conversion model, transmitter-side modulation design, and passive wing-tag design. The model explains the energy conversion chain from optical absorption to photothermal heating and thermal-to-mechanical vibration, guiding both signal processing and hardware design. Based on this model, we design the transmitter to produce optical excitation through tailored light signal and modulation strategy, enabling different target signals to be embedded for different applications. In parallel, we develop a practical butterfly wing tag by selecting suitable species and wing regions and designing lightweight stabilization structures that improve robustness without compromising photoacoustic efficiency, enabling flexible and even wearable deployment.

Together, these model-guided designs transform the photoacoustic effect from a specialized, costly system into a low-cost, end-to-end platform that can be assembled from

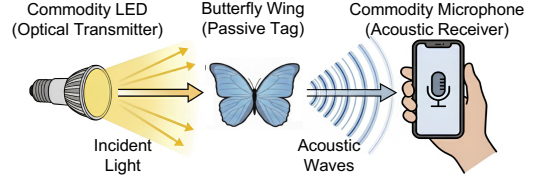


Figure 3: Proposed butterfly photoacoustic system.

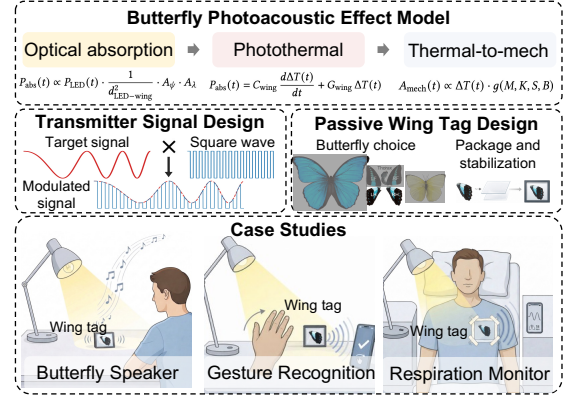


Figure 4: System overview.

everyday components. We demonstrate a typical desktop illumination scenario in which an ordinary LED desk lamp illuminates a nearby passive butterfly tag. Under this setup, we demonstrate direct audible sound playback using butterfly tags, enabling interactive audio experiences without requiring any electronics on the user—only a single butterfly wing. In the ultrasonic band, we demonstrate two sensing tasks: large-motion hand gesture recognition and small-motion respiration monitoring, using a butterfly wing as a passive tag. The primary goal of this work is to characterize the butterfly photoacoustic effect and establish a platform for the research community, while leaving broader application exploration to future work. In the following sections, we first present the energy conversion model and then the transmitter and tag designs.

4 Modeling Butterfly Photoacoustic Effect

Based on prior photoacoustic studies [29, 30, 65] and our comprehensive experiments, we develop a qualitative model tailored to butterfly wings to capture key insights into the light-heat-acoustic conversion process and guide the subsequent system design.

4.1 Deriving the Energy Conversion Model

Stage I: optical absorption. The first stage of the butterfly photoacoustic effect is optical absorption, which determines how much incident optical power is captured by the wing and transferred to subsequent thermal and mechanical conversion processes. Accordingly, the absorbed optical power [27]

$P_{\text{abs}}(t)$ can be modeled as follows:

$$P_{\text{abs}}(t) \propto \frac{1}{d_{\text{LED-wing}}^2} \cdot A_{\psi} \cdot \eta_{\text{wing}} \cdot P_{\text{LED}}(t), \quad (1)$$

where $d_{\text{LED-wing}}$ is the distance between the LED and butterfly wing, A_{ψ} represents the loss caused by the incident angle ψ , η_{wing} is the optical absorption coefficient of the butterfly wing, and $P_{\text{LED}}(t)$ is the LED's instantaneous optical power.

In this model, $P_{\text{LED}}(t)$, $d_{\text{LED-wing}}$, and A_{ψ} mainly capture the effect of the light source and the relative geometry between the LED and the wing, while η_{wing} reflects the wing's intrinsic ability to absorb incident light. η_{wing} is mainly determined by the wing's structural and material properties. Structurally, a butterfly wing contains numerous ridges that act as remarkable photonic nanostructures [44]. Although the exact ridge sizes and the spacing between neighboring ridges vary across species, they are typically on the order of hundreds of nanometers to a few micrometers [34, 50], comparable to visible-light wavelengths (400–700 nm). Thus, light of different wavelengths interacts differently with the wing structure: light at some wavelengths is more effectively confined within the ridges and undergoes a greater number of reflections, enhancing absorption. Material-wise, the wing is primarily made of chitin, a material that is highly effective at absorbing optical energy when combined with the nanostructure [37]. Together, these structural and material features render butterfly wings highly effective optical absorbers.

However, optical absorption alone does not determine the final photoacoustic output. The absorbed energy must still be converted into sufficiently strong temperature oscillation and subsequent mechanical vibration, which are further constrained by the thermal and mechanical stages below.

Stage II: photothermal conversion. After optical absorption, the butterfly wing converts the absorbed optical energy into heat, so that it heats up when the absorbed optical energy increases and cools down when the energy diminishes. To characterize this heating-cooling process, we model the wing as a first-order thermal system [14, 57]:

$$P_{\text{abs}}(t) = C_{\text{wing}} \frac{d\Delta T(t)}{dt} + G_{\text{wing}} \Delta T(t), \quad (2)$$

where $\Delta T(t)$ is the temperature variation, C_{wing} is heat capacity of the heated region, and G_{wing} is heat loss coefficient.

Equation 2 shows that the temperature response is jointly determined by the absorbed optical power $P_{\text{abs}}(t)$ and the wing's thermal properties (C_{wing} and G_{wing}). This stage depends not only on the amount of optical energy delivered, but also on the rate (power) at which it is delivered. Since the photoacoustic effect relies on periodic temperature fluctuation rather than merely a high average temperature, strong photoacoustic output requires the wing temperature to rise and fall both sufficiently in magnitude and rapidly within

each modulation cycle. This directly leads to two requirements on optical excitation. First, the optical signal should have sufficiently sharp ON/OFF transitions. Gradual intensity changes smooth the temperature variation, weakening the thermoelastic response. Second, the light ON/OFF durations must be properly designed. If the duration is too short, the wing cannot heat or cool sufficiently before the next cycle, leading to weaker temperature oscillations.

Butterfly wings are ultra-thin and low-density, leading to a small heat capacity C_{wing} and allowing a noticeable temperature rise with a small amount of absorbed optical energy, which is favorable for photothermal conversion. Meanwhile, their porous chitin-air architecture and exposed surface facilitate heat dissipation to the surrounding air, corresponding to a relatively large heat loss coefficient G_{wing} . These properties allow the wing temperature to rise and fall quickly.

Stage III: thermal-to-mechanical conversion. In this stage, the temperature fluctuation generated in Stage II drives the mechanical vibration of the butterfly wing. This process is essentially governed by the thermoelastic effect [54]: temperature variation $\Delta T(t)$ causes repeated expansion and contraction of the wing, which then drives its vibration. However, the final vibration strength is not determined by temperature variation alone. After the thermoelastic deformation, the actual vibration response further depends on the mechanical properties of the wing. Based on the mechanical vibration theory [36], even under the same driving input, differences in structure and material properties can lead to different vibration responses. Here, we model the butterfly wing as an ultra-thin, lightweight vibrating membrane and capture the dominant factors governing its vibration with a simplified relation:

$$A_{\text{mech}}(t) \propto \Delta T(t) \cdot g(M, K, S, B), \quad (3)$$

where $A_{\text{mech}}(t)$ denotes the vibration strength, M is the mass of the vibrating wing, K is the mechanical stiffness, S is the vibrating area, and B is the boundary condition. This equation shows that the temperature variation $\Delta T(t)$ determines the strength of the thermoelastic driving, while $g(M, K, S, B)$ describes butterfly wing's efficiency in converting this driving into actual vibration. Even under similar thermal excitation, different butterfly species, wing regions, and mounting conditions can lead to varying mechanical responses.

The above analysis also explains why butterfly wings are well suited to our system. Their ultra-thin, lightweight chitin-based membranes, supported by a vein network, combine low mechanical inertia with a deformable vibrating surface. Together, these features make the wing highly responsive to thermoelastic driving, favoring efficient vibration generation. At the same time, differences in mass, stiffness, and vibrating area across species and wing regions lead to varying vibration responses. In general, heavier or stiffer regions

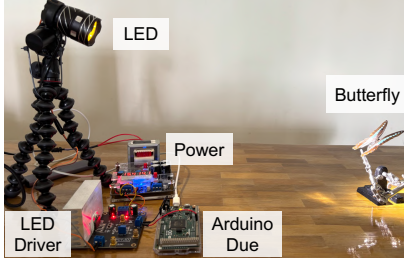


Figure 5: Experiment setup for model validation.

are expected to deform less under the same thermoelastic driving, producing weaker vibrations, whereas regions with larger vibrating areas can generate stronger acoustic output.

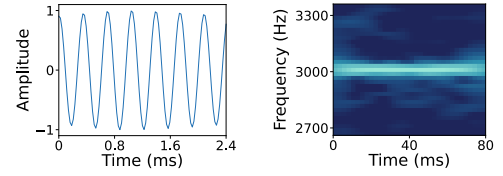
Among these factors, the boundary condition is particularly important. When the wing remains attached to the intact butterfly, its vibration is mechanically constrained by the butterfly body. In contrast, an isolated wing provides a simpler and more stable boundary condition, making it more suitable for efficient photoacoustic generation.

4.2 Model Validation

Experiment setup. The setup is shown in Figure 5. At the optical transmitter, we use an Arduino DUE to generate the signal for an LED driver (OPA549 [47]), which drives a commercial LED (white Cree XLamp XHP70 [10], 15 W). We first use the sine wave as the default modulation signal for model validation, and then evaluate different waveforms. Figure 6 shows the wing-generated acoustic waveform and spectrogram under a 3 kHz sinusoidal modulation, illustrating the fidelity of the generated acoustic signal. For the butterfly tag, we first use an intact, naturally deceased butterfly (*Graphium sarpedon* [55]) suspended in air. We also test other species to examine cross-species consistency. For factors exhibiting similar trends, such as LED–wing distance, we report the representative result obtained from *Graphium sarpedon*. For species-dependent factors, such as LED color and wavelength, we conduct additional experiments across species. Here, the wings are not detached from the body, and no transparent films are added to stabilize the wing. The LED–wing distance is fixed at 30 cm. We use a commercial smartphone (Samsung S21 Ultra [41]) as the acoustic receiver to record the sounds generated by the wing, with the phone placed 2 cm away from the wing. We adopt the widely used Sound Pressure Level (SPL) as the primary evaluation metric.

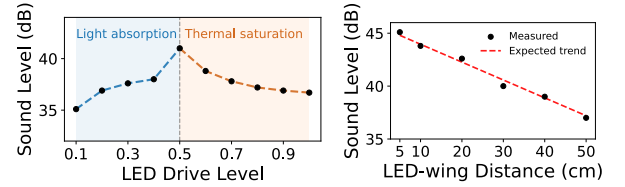
Optical absorption. We validate the optical absorption model in Equation 1 by isolating each factor and assessing its end-to-end impact on the resulting acoustic output.

Emitted LED power. We vary the LED power level while keeping all other parameters fixed. As shown in Figure 7(a), SPL first increases with LED power, but gradually saturates and slightly decreases at higher power. The initial increase

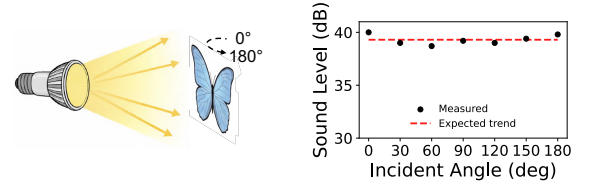


(a) Acoustic waveform (b) Acoustic spectrogram

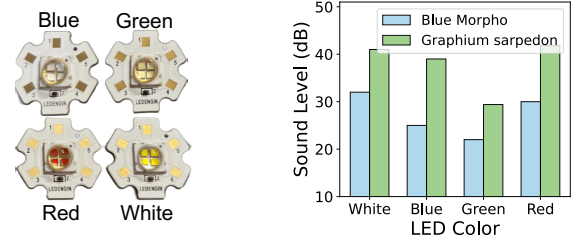
Figure 6: Acoustic output generated by the wing.



(a) Impact of incident LED power (b) Impact of LED-wing distance



(c) Incident angle setup (d) Impact of LED incident angle



(e) Different color LED (f) Impact of LED wavelength

Figure 7: Validation of Stage-I optical factors.

is consistent with Equation 1, since higher $P_{LED}(t)$ leads to larger absorbed optical power. However, at even higher power, the accumulated heat becomes excessive and cannot dissipate during the cool-down period. This weakens the temperature rise–fall pattern, causing the output to stop increasing and instead decline.

LED–wing distance. We vary the LED–wing distance from 5 cm to 50 cm while keeping other parameters unchanged. Figure 7(b) shows that the measured SPL decreases as the LED–wing distance increases. This is because less light energy reaches the wing at a longer distance, leading to reduced absorbed optical power, which well matches the model.

Incident angle. We rotate the wing relative to the LED to vary the incident angle (Figure 7(c)) while keeping other parameters unchanged. As shown in Figure 7(d), the SPL changes only slightly, indicating weak angle dependence of

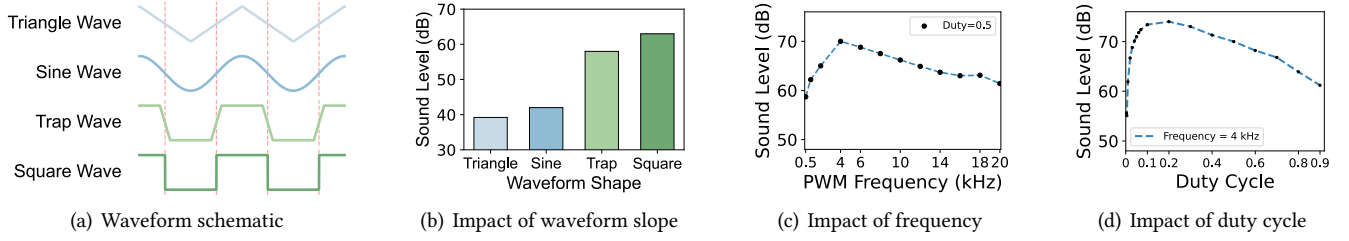


Figure 8: Validation of Stage-II photothermal conversion.

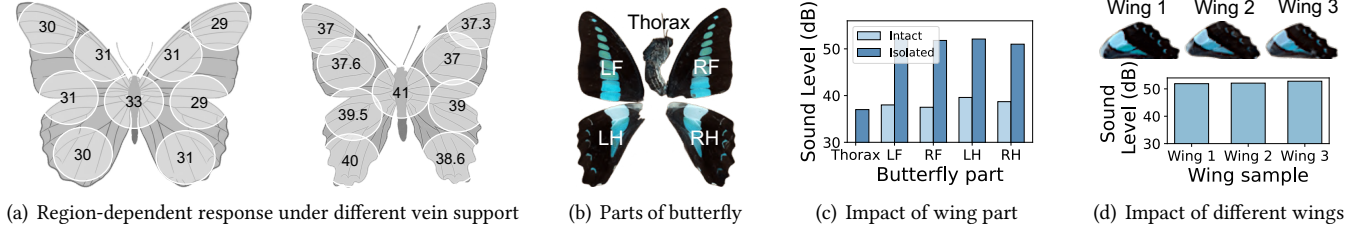


Figure 9: Validation of Stage-III mechanical vibration factors.

the butterfly photoacoustic effect. This is because the LED provides broad, uniform illumination across the wing, so wing rotation has little effect on the incident optical energy.

Wavelength. We conduct experiments under different illumination wavelengths (Figure 7(e)) and compare two butterfly species under the same conditions. Figure 7(f) shows clear wavelength dependence in the photoacoustic response, which is consistent with the wavelength-dependent optical absorption model developed in Stage I. For *Graphium sarpedon*, green illumination yields noticeably lower sound than white, blue, and red illumination, consistent with its greenish appearance: more green light is reflected and less is absorbed for photoacoustic conversion. A similar wavelength selectivity is also observed for Blue Morpho. Commonly used white light yields relatively strong output for both species, because its broadband spectrum still provides substantial absorbed energy even if part of the spectrum is reflected.

Photothermal conversion. Equation 2 models the wing's photothermal heating-cooling process as a first-order thermal system. Since the underlying thermal parameters are difficult to control directly, we validate the model by varying the optical modulation, which controls the time-varying absorbed energy and thermal response, and examining the resulting photoacoustic output.

Transition slope (waveform shape). To examine the impact of optical transition sharpness, we compare optical modulation waveforms with increasingly sharp edges (triangle, sine, trapezoid, and square), while keeping other parameters fixed (Figure 8(a)). As shown in Figure 8(b), sharper transitions consistently produce stronger photoacoustic output, with square-wave modulation achieving the highest sound

level. This confirms that waveform slope directly affects the thermal driving strength: slowly varying waveforms lead to weaker temperature variation, whereas sharper transitions induce stronger thermoelastic excitation.

Modulation frequency. We vary the square-wave modulation frequency to change the duration of the cycle while fixing the duty cycle at $D = 0.5$, i.e., with equal light ON/OFF durations within each cycle, and keep the other factors unchanged. Figure 8(c) shows a clear unimodal response: the output first increases and then decreases with frequency. At low modulation frequencies, each heating-cooling cycle is long, so the wing expands and contracts too slowly to generate strong acoustic waves. As the frequency increases, the expansion and contraction become faster, which enhances the acoustic output. When the frequency becomes too high, however, the modulation period becomes too short relative to the wing's thermal response time, so the wing can neither heat up enough nor cool down enough within each cycle, leading to reduced photoacoustic output.

Duty cycle. We now fix the modulation frequency at 4 kHz, which gives strong performance in the previous experiment, and then vary the duty cycle to examine its impact. As shown in Figure 8(d), the photoacoustic output varies across different duty cycles, with the strongest response appearing when the duty cycle is around 0.2. This trend is consistent with the Stage-II model: the butterfly wing can heat up quickly under illumination, but it needs more time to dissipate heat. This result shows that at a particular modulation frequency, the duty cycle remains an important factor affecting photoacoustic output. Note that the frequency and duty cycle must be jointly searched to achieve the best performance.

Thermal-to-mechanical conversion. To validate Stage III, we vary the conditions in Equation 3 and measure the resulting photoacoustic output.

Region-dependent mechanical response. We shine the same small light spot on different regions of the wing and compare the resulting sound levels. As shown in Figure 9(a), both Blue Morpho (left) and Graphium sarpedon (right) exhibit regional variations in performance. Positions closer to the wing root and major supporting veins tend to produce stronger photoacoustic output, while regions near the outer edge produce slightly weaker output. This suggests that local mechanical support and stiffness distribution affect the wing's vibration and sound generation, although the overall variation across regions remains small.

Whole body vs. detached wing. To assess the contributions of different butterfly parts (Figure 9(b)) to the photoacoustic effect, we measured the intact butterfly, the detached wing, and the isolated thorax. As shown in Figure 9(c), all detached wings generate clearly higher sound levels than when the same wings remain attached to the body, whereas the thorax alone produces only weak output. This indicates that the body primarily adds mechanical loading and coupling, rather than contributing to acoustic generation.

Wings from different individuals of the same species. We further compare the same wing region across individuals of the same species. The similar SPLs in Figure 9(d) indicate that within-species variation has little effect on photoacoustic output, demonstrating good repeatability.

5 System Design

5.1 Transmitter Modulation Design

Guided by the model and validation results in Section 4, we optimize the transmitter modulation to generate strong and stable photoacoustic output.

Waveform shape. Guided by our model, we use a square-wave signal instead of a sinusoidal one, since its sharper ON/OFF transitions produce stronger thermal excitation. This increases the acoustic output from about 40 dB to around 60 dB (Figure 8(a)-(b)).

Absolute heating and cooling time. Beyond waveform shape, our model further shows that the absolute heating and cooling durations within each cycle are also critical. We thus jointly tune the square-wave modulation frequency (cycle duration) and duty cycle, further increasing the photoacoustic output to around 70 dB.

Carrier-based envelope modulation. With the above design, we can achieve a strong photoacoustic output under a fixed modulation frequency. In practical audio playback, the signal frequency changes over time. If we directly use square-wave modulation for the target audio, two problems arise. First, the generated photoacoustic signal's amplitude

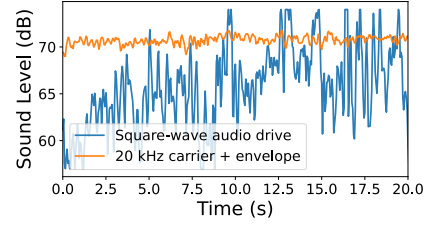


Figure 10: Square-wave vs. carrier-based modulation.

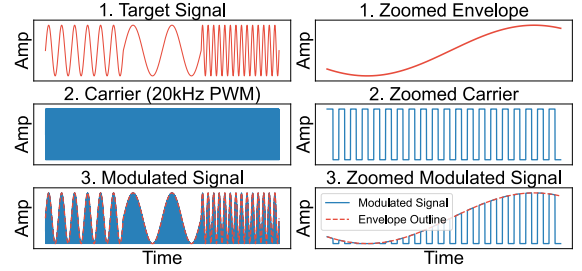


Figure 11: Carrier-based modulation schematic.

fluctuates dramatically, as shown in Figure 10. This is because different audio frequencies correspond to different cycle durations, which influence the heating-cooling intervals and, consequently, the acoustic amplitude. Second, directly encoding audio using square waves introduces noticeable waveform distortion, which degrades audio quality.

To address these issues, we adopt a fixed high-frequency carrier (e.g., 20 kHz) with the optimized duty cycle of 40%, and modulate the target audio signal onto the carrier as its envelope, as illustrated in Figure 11. Formally, the modulated optical signal is $M(t) = A_{\text{audio}}(t) \cdot C(t)$, where $C(t) \in \{0, 1\}$ is a fixed-frequency square-wave carrier and $A_{\text{audio}}(t)$ is the audio envelope. This design offers two key benefits. First, the fixed carrier frequency maintains consistent light ON/OFF durations, ensuring a stable photoacoustic output. Second, encoding the target audio in the envelope better preserves its temporal variations while reducing waveform distortion.

An interesting result is that the generated photoacoustic output contains both a high-frequency carrier component and a low-frequency audio (envelope) component. However, because the carrier lies in the ultrasonic range, it is inaudible to the human ear, eliminating the need for demodulation during audio playback. This property enables the butterfly-wing tag to operate without any electronic components.

5.2 Butterfly Wing Tag Design

The tag design aims to enhance both signal strength and quality through species and region selection, as well as stabilization to suppress unwanted nonlinear components.

Species and wing-region selection. Although there are tens of thousands of butterfly species in nature, only

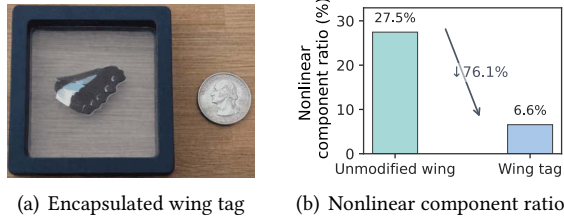


Figure 12: Tag prototype and nonlinear reduction.

a few dozen are widely bred commercially [5]. Guided by our model, we derive three practical guidelines for selecting species and wings. (i) Butterfly species should be chosen to match the light source. For example, yellow or white wings tend to reflect more light and produce weaker photoacoustic output, whereas darker species, such as *Graphium sarpedon*, are generally more favorable. (ii) The selected wing region should provide favorable mechanical support. Regions near the wing root or major veins tend to produce stronger output. Detached wings outperform wings attached to the body. In our experiments, the default tag is a single wing of roughly coin size (Figure 12(a)). (iii) Whenever possible, low-cost and widely available butterfly species should be chosen. Our evaluation of 23 commonly bred species shows that strong photoacoustic performance does not require rare species.

Encapsulation and stabilization. Even though the design at the transmitter side improves signal strength and stability, the generated photoacoustic signals still contain irregular nonlinear spectral components. These components persist even under single-frequency sinusoidal excitation, suggesting that they primarily originate from the butterfly wing itself. This is because butterfly wings are ultra-light and highly sensitive to external disturbances, such as airflow and unstable boundary conditions.

A straightforward solution is to use heavier butterfly species or add extra load to the wing. However, these approaches perform poorly in practice: heavier wings exhibit weaker vibrational responses, while additional loading introduces excessive damping, severely degrading the photoacoustic output. An effective stabilization design should remain optically transparent while introducing only minimal mechanical constraints. Guided by this principle, we sandwich the wing between two ultra-thin transparent films and fix it at the edge, as shown in Figure 12(a). This design suppresses irregular vibration while minimally influencing optical absorption and thermo-mechanical vibration, and also protects the wing. For wearable or flexible deployments, the rigid frame can be replaced by other edge-fixing methods, such as tape.

We evaluated three representative encapsulation materials: polyvinyl chloride (PVC) [32], polydimethylsiloxane (PDMS) [2], and polyethylene (PE) [38]. PVC is too stiff and introduces strong damping, while PDMS softens and deforms

under heating, leading to unstable responses. PE provides the best balance of low added mass, optical transparency, thermal stability, and mechanical compliance, and is therefore chosen as the encapsulation material. With PE encapsulation, the nonlinear component ratio, defined as the fraction of spectral energy outside the target frequency, is significantly reduced from 27.5% to 6.6%, while the photoacoustic output strength decreases by only 1.7% (Figure 12(b)). These results show that proper encapsulation improves signal quality while preserving nearly all useful photoacoustic output.

6 Implementation

Optical transmitter. The optical transmitter, shown in Figure 13(a), consists of an Arduino Due (\$36) that controls an LED driver (OPA-549 [47], \$28) to drive a commercial white LED (Cree XLamp XHP70 [10], 15 W, \$2).¹ For carrier-based envelope modulation, the Arduino Due synchronously outputs the high-frequency square-wave carrier $C(t)$ from a digital pin and the audio envelope $A_{\text{audio}}(t)$ from its DAC. A low-cost analog switch board (CD74HC4067 [48], \$1.6) gates the envelope with the carrier, implementing $M(t) = A_{\text{audio}}(t) \cdot C(t)$ in hardware while preserving sharp carrier edges. Before measurement, we calibrate the incident light using a photodiode, disable AGC and filtering, and validate the acoustic output against a MEMS reference microphone.

Butterfly-wing tag. We use farmed, naturally deceased butterfly specimens throughout our experiments. We adopt the tag design described in Section 5.2: an isolated wing is encapsulated between two ultra-thin transparent PE films and further stabilized as shown in Figure 12(a) and Figure 13(a). Unless otherwise specified, our default tag is made of the lower wing of *Graphium sarpedon*, a common and low-cost species, with each wing costing less than \$0.5. The encapsulation films and the peripheral frame are also inexpensive and readily available, making the total material cost of a tag below \$1. To evaluate generality, we further test the proposed system across 23 butterfly species, as shown in Figure 13(b).

Baselines and comparison objects. To benchmark the butterfly wing against other lightweight and practically accessible structures, we use four representative baselines, as shown in Figure 13(c). The piezoelectric diaphragm has been used in prior photoacoustic sensing systems [35]. Stethoscope diaphragm and dragonfly wings are also selected for comparison with butterfly wings. The former is thin and readily vibrated, while the latter features light-trapping microstructures but differs from butterfly wings in optical, material, and mechanical properties. We also include a smartphone microphone, as prior work [45] has demonstrated light-induced photoacoustic responses in MEMS microphones under modulated laser illumination.

¹The typical power range of common LED desk lamps is 5–20 W [4, 16].

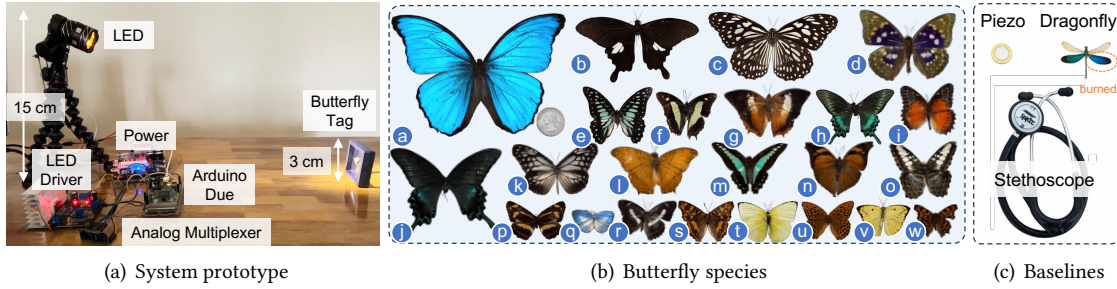


Figure 13: System prototype, butterfly species evaluated, and baseline comparison objects.

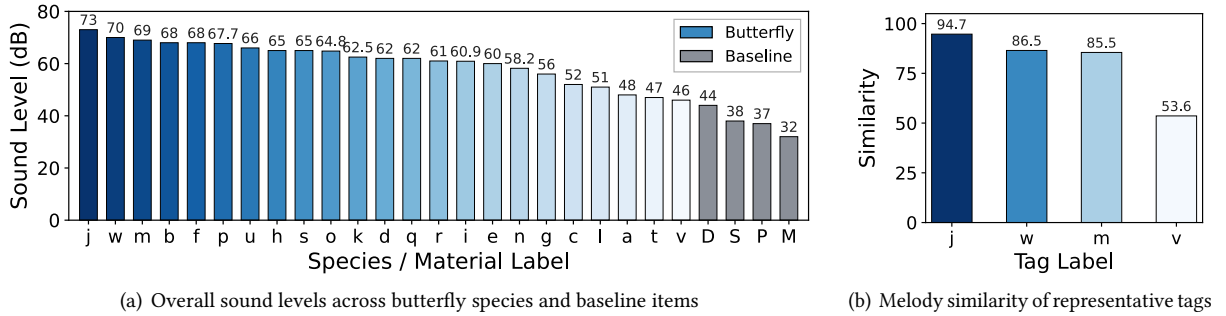


Figure 14: Overall performance and generality of butterfly wing tags.

7 Evaluation

7.1 Overall Performance and Generality

Generality across butterfly species. We evaluated the performance across 23 butterfly species. All species are tested under the same configuration, including white LED illumination and square-wave modulation, to ensure a fair cross-species comparison rather than a per-species optimized upper bound. As shown in Figure 14(a), most butterfly tags generate strong acoustic outputs, and many exceed 60 dB, with the best-performing samples reaching 65–73 dB. This result shows that strong photoacoustic emission is not limited to a single exceptional specimen, but can be achieved across a broad range of commonly seen butterfly species.

Comparison with baseline objects. We further compared butterfly wing tags with the four baseline objects introduced in Section 6: a dragonfly wing (D), a stethoscope diaphragm (S), a piezoelectric diaphragm (P), and a smartphone microphone (M). Since all objects are tested under the same illumination setting, the baseline comparison in Figure 14(a) also provides relative evidence of photoacoustic energy-conversion efficiency. The sound level generated by the butterfly wing tag reaches up to 73 dB, compared with 44 dB, 38 dB, 37 dB, and 32 dB for the dragonfly wing, stethoscope diaphragm, piezoelectric diaphragm, and microphone diaphragm, respectively. These differences correspond to roughly 800×, 3200×, 4000×, and 12,600× higher acoustic intensities, respectively.

Melody similarity of representative tags. We further evaluate melody reproduction using a note-level similarity score. For each butterfly tag, we extract the detected note sequence and compare it with the reference melody using dynamic time warping (DTW), a commonly used method for melody matching under timing variations [49]. DTW compensates for small timing offsets caused by note-boundary shifts and slight duration variations. The final score is computed as $\text{Score} = 100 \exp(-1.5\bar{c})$, where $\bar{c}$ is the average normalized note-index difference. A score of 100 indicates a perfect match, while lower scores indicate greater note mismatches. As shown in Figure 14(b), butterfly tags with stronger acoustic outputs generally achieve higher melody similarity, whereas weaker tags tend to achieve lower scores due to their reduced SNR. Overall, most butterfly wing tags can accurately reproduce the melodies.

7.2 Impact of System Parameters

Impact of carrier frequency. We varied the ultrasonic carrier frequency from 18 kHz to 22 kHz while keeping other configurations the same. As shown in Figure 15(a), the output sound level is similar across different carrier frequencies. Figure 15(b) further shows that the melody similarity remains high across all tested carrier frequencies. These results indicate that the choice of carrier frequency has a negligible impact on the quality of the generated acoustic signals. We therefore use 20 kHz as the default carrier frequency in the remaining experiments.

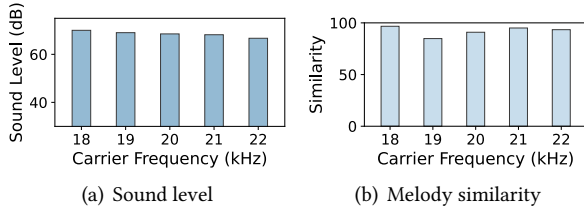


Figure 15: Impact of carrier frequency.

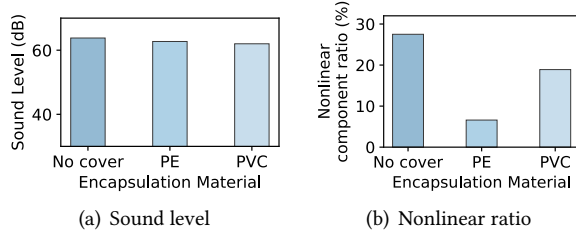


Figure 16: Impact of encapsulation material.

Impact of encapsulation material. We compared three encapsulation conditions: no cover, PE film, and PVC film. As shown in Figure 16(a), encapsulation causes only a small reduction in output sound level, with PE reducing it by just 1.7% and preserving a higher sound level than PVC. Figure 16(b) shows a much clearer difference in spectral quality measured by the nonlinear component ratio. PE yields the lowest ratio (6.6%), PVC gives a higher ratio (18.9%), and the uncovered wing exhibits the highest ratio (27.5%). These results show that proper encapsulation suppresses irregular acoustic components and improves spectral cleanliness, with PE outperforming PVC in both signal preservation and nonlinear distortion reduction.

7.3 Robustness Evaluation

Long-term stability. We evaluated the stability of the butterfly wing tag over a continuous 30-minute run by measuring the output sound level every 2 minutes. As shown in Figure 17(a), the output remains highly stable, with only around 0.5 dB fluctuation throughout the test. Beyond this continuous test, several PE-encapsulated butterfly tags in our lab have been retained and reused for more than 12 months without noticeable structural failure or performance degradation. The PE film helps protect the fragile wing during handling and use, while preserved butterfly specimens have been documented to remain intact for decades in museum collections [28]. Together, these observations provide evidence of the practical long-term stability and durability of the proposed tag design.

Impact of ambient light. We further tested the system under different ambient lighting conditions, including morning, noon, night (dark), and night with a nearby desk lamp. As shown in Figure 17(b), the measured sound level is similar

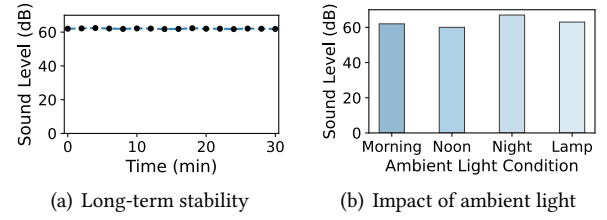


Figure 17: Robustness and reliability of wing tags.

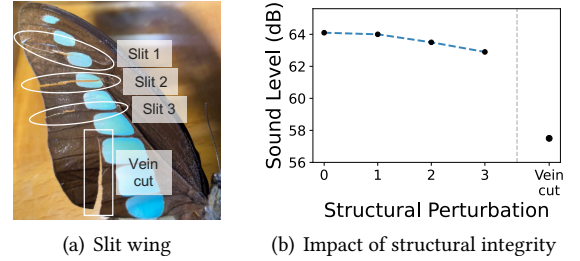


Figure 18: Impact of the wing's structural integrity.

under all conditions. This result shows that the system is generally robust against common variations in ambient light.

Damage tolerance. In practice, butterfly specimens often arrive with minor natural imperfections, such as local scale loss or slight edge damage. These small defects usually have only a limited impact on the photoacoustic response. To systematically evaluate damage tolerance, we cut the wing, as shown in Figure 18(a). We first create 1–3 slits in the membrane region while keeping the major veins intact. As shown in Figure 18(b), the sound level decreases slightly, from 64.1 dB without cuts to 62.9 dB with three slits. This suggests that slight membrane damage is tolerable. By contrast, cutting a primary supporting vein causes a much larger drop in sound level, down to 57.5 dB, highlighting the important role of the vein network in the wing's mechanical response.

7.4 Case Studies

We demonstrate the practical potential of the butterfly photoacoustic effect through two case studies: a butterfly loudspeaker for direct audible audio delivery, and a passive ultrasonic sensing platform for detecting both large motions (e.g., hand gestures) and subtle motions (e.g., respiration).

7.4.1 Butterfly photoacoustic speaker. The butterfly photoacoustic speaker shows that a single passive butterfly wing can directly generate audible sound without any electronics, battery, or wireless module on the tag side. This makes it attractive for simple audio-delivery applications such as interactive museum guides, where a visitor only needs to wear or approach a butterfly tag to hear the audio.

Wing-microphone distance. We varied the distance between the butterfly wing and the microphone while keeping other conditions unchanged. As shown in Figure 19(a), the

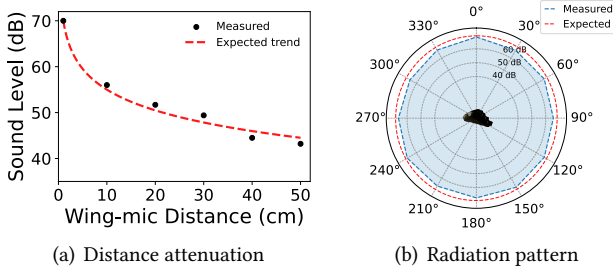


Figure 19: Acoustic propagation characteristics.

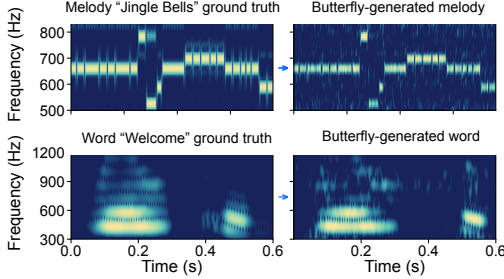


Figure 20: Spectra of melody and voice-prompt.

received sound level decreases as the distance increases. This is expected, since the sound becomes weaker as it propagates through air. The results provide a reference for the practical playback range of the butterfly speaker. The threshold of hearing for a healthy human ear is approximately 0 dB SPL, and in our experiments, the sound can still be perceived at a distance of 2 m. Notably, this range is achieved without any external acoustic structure. The transmission range could be further extended by introducing mechanical structures to confine and direct the sound in a particular direction.

Radiation direction. We rotated the microphone around the butterfly wing at a fixed distance. As shown in Figure 19(b), the sound level varies little across angles, indicating nearly omnidirectional radiation.

Audio playback examples. We present the reference and generated audio samples (“Jingle Bells” and “Welcome”) in the time-frequency domain in Figure 20. The melodies achieve 96.3% average note-level similarity, while 100 common spoken words achieve an average STOI of 0.56. We observe that preserving speech fidelity is more challenging than reproducing melodies because speech spans a wider frequency range and contains more diverse frequency components. Because butterfly wings respond differently across frequencies, speech components are reproduced at different intensities, resulting in spectral distortion. Potential solutions include selecting wing regions that better support the target speech frequency range and designing frequency-selective algorithms to preserve important speech components while suppressing less important components when the wing cannot reproduce all frequency components at the

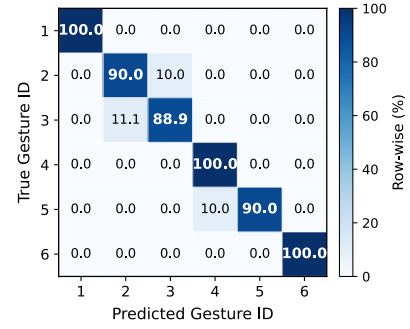


Figure 21: Gesture recognition confusion matrix.

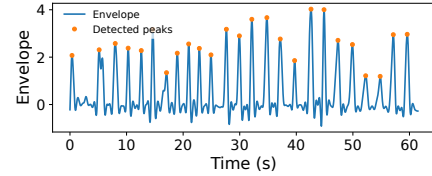


Figure 22: Sensed respiration waveform.

same intensity. Demo videos and audio samples are available at <https://github.com/lupengz/ButterflyEffect>.

7.4.2 Hand gesture recognition. We demonstrate an ultrasound sensing case for hand gesture recognition. In this setup, an LED illuminates the butterfly wing with an 18 kHz modulated light signal, and the wing generates a corresponding ultrasonic photoacoustic signal. A commodity smartphone microphone records the signal while the user performs a set of standard hand gestures [56] between the LED and the tag, without touching any electronics. Different gestures block the incident light differently, leading to distinguishable acoustic patterns. Figure 21 shows the row-wise normalized confusion matrix. Using a Manhattan-distance classifier, we achieve 94.7% accuracy across six gestures with 20 repetitions per gesture. Most errors occur between gestures with similar light-blocking patterns. This demonstrates fine-grained, battery-free, non-contact gesture sensing using only an LED, a commodity microphone, and a butterfly.

7.4.3 Respiration monitoring. We next demonstrate a small-motion sensing case through respiration monitoring. In this setup, we replaced the rigid holder with tape and attached the butterfly wing tag directly onto the subject’s clothing near the chest. An LED illuminates the wing with an 18 kHz modulated light signal; a smartphone microphone records the resulting ultrasonic photoacoustic signal. During breathing, chest motion slightly moves the tag and varies the received acoustic amplitude, producing a clear waveform that follows the respiration pattern. We extracted the signal envelope and detected peaks to count breathing cycles. As shown in Figure 22, the extracted waveform clearly follows respiration. Across 20 repeated trials, the average respiration-rate error

is only 0.93 breaths per minute. These results show that a butterfly wing can serve as a lightweight and unobtrusive wearable for respiration monitoring.

8 Discussions

Ethics and sourcing. All butterfly wings used in this study were obtained from commercially bred, naturally deceased specimens rather than wild-caught butterflies. Such specimens are already supplied by butterfly farms for crafts, jewelry, and educational purposes [5]. Our work therefore does not require wild collection or harming butterflies. The sub-\$1 tag cost reflects the low material cost rather than encouraging disposable use. Moreover, PE encapsulation improves tag durability and reduces the need for frequent replacement, supporting more sustainable long-term use.

More applications. The proposed system has the potential to support a broader range of applications. Its lightweight, battery-free, metal-free, and magnet-free tag design is well suited to restricted environments, such as hospitals and industrial settings, where electronics or metals may be undesirable. It also enables low-power, low-waste sensing. For example, butterfly wings could support gas sensing, as gas molecules permeating the wing nanostructures may alter optical absorption and produce distinguishable photoacoustic spectra for detection and identification.

Signal manipulation. In this work, the generated acoustic signal is used as-is, without any manipulation. External structures can be applied to confine the sound in the desired direction, thereby increasing the strength and range of the signal. Multiple light sources can also be used to illuminate the same tag from different directions, increasing the delivered energy and reducing sensitivity to orientation.

9 Related Work

Photoacoustic effect. The photoacoustic effect has been widely studied in medical imaging and spectroscopy [40, 52], but such systems often require high-end light sources, sensitive acoustic transducers, and careful alignment/calibration [50, 51, 64], limiting them to specialized settings. A few works explore more practical uses: Nutrilizer [35] uses a 16-LED array and piezoelectric sensors for liquid sensing, while LightCommands [45] uses lasers to inject fake audio commands into voice assistants. In contrast, our work uses low-cost off-the-shelf components and ordinary butterfly specimens to generate strong photoacoustic signals, moving photoacoustic systems closer to everyday applications.

Butterfly-wing-based systems. Butterfly wings have been widely studied in optics, where their natural nanostructures have inspired optical reflectors and light-trapping systems [9, 15, 46]. Their lightweight structure has also been used for speech recovery: prior work [66] directs a laser at a

butterfly wing to measure speech-induced wing vibrations for speech recovery. In that system, the wing acts as a passive acoustic receiver. In contrast, our system exploits the photoacoustic effect: modulated light drives the wing to vibrate and actively generate designed acoustic waves, making the wing a light receiver and an acoustic transmitter. The underlying mechanism and system goal are completely different.

Thin and flexible speakers. Prior works have explored alternatives to conventional magnetic loudspeakers, such as piezoelectric transducers, electrostatic speakers, and thermoacoustic thin-film speakers [20, 24, 57]. Although these designs can be thin or flexible, they still require electrical actuation and driving circuits [33]. The proposed tag operates without an onboard power supply, electronics, or magnets, and can be activated using an everyday LED.

Wireless sensing beyond RF. While Radio Frequency (RF)-based sensing (e.g., WiFi [23, 58, 60, 63], RFID [42, 59], and mmWave [25, 61]) has been extensively studied, non-RF sensing remains important when RF is limited, susceptible to interference, or unsuitable for EMI-sensitive environments [6, 13]. Major non-RF alternatives include visible-light sensing [1, 11, 17, 26] and acoustic sensing [7, 43, 62]. Unlike prior work that uses either modality alone, our work couples light and sound through photoacoustic energy conversion: light actuates a fully passive butterfly tag, and the generated sound supports information delivery and sensing. This creates a new cross-modal non-RF sensing modality and can inspire further system designs and applications.

10 Conclusion

We present the first butterfly photoacoustic system using natural butterfly wings and develop a model of the photoacoustic process to lay the foundation for future studies. Leveraging this model, we design both the optical transmitter and butterfly-wing tag, enabling the photoacoustic effect with low-cost, off-the-shelf hardware. Our system produces audible sound above 70 dB, enabling a bio-speaker, while also supporting fine-grained ultrasonic sensing with ordinary butterfly wings. We envision that this work will inspire further research into bio-inspired designs and applications.

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